



## **Laboratorio SMTP – POP3 - IMAP**

**Bernardo Rizzone (232013194)**

No laboratório a seguir eu apresentei algumas dificuldades. Todas as partes 1 a 3 eu consegui concluir normalmente obtendo base 64 do meu email e da senha app password. Consegui abrir conexão segura com o SMTP através do openssl e consegui fazer o handshake SMTP + AUTH LOGIN. Porém, na parte de enviar a mensagem eu não consegui de forma alguma passar do RCPT TO. Apresentando erros em todas as soluções dadas no PDF, como trocar a porta para 465 (TLS IMPLÍCITO) e trocar - TLS1\_2. Todas as formas deram erro no RCPT TO. Então optei por fazer com a ferramenta swaks. Obtendo assim êxito na tarefa. Vou deixar em anexo agora o print dos erros e depois do uso swaks.

**Imagem 1.** Nessa imagem foi o erro obtido usando o código normal proposto.

```
MAIL FROM:<bernardorizzone99@gmail.com>
250 2.1.0 OK d9443c01a7336-29498d23226sm1003865ad.49 - gsmt
RCPT TO:<bernardorizzone98@gmail.com>
RENEGOTIATING
40A73D1F827D0000:error:0A00010A:SSL routines:can_renegotiate
:wrong ssl version:../ssl/ssl_lib.c:2323:
```

**Imagem 2.** Nessa imagem foi o erro obtido usando o código com alteração no TLS1\_2.

```
MAIL FROM:<bernardorizzone99@gmail.com>
250 2.1.0 OK 98e67ed59e1d1-33fed73a7b7sm42733a91.5 - gsmt
RCPT TO:<bernardorizzone98@gmail.com>
RENEGOTIATING
4007DD2988790000:error:0A00044C:SSL routines:ssl3_read_bytes
:tlsv1 alert no renegotiation:../ssl/record/rec_layer_s3.c:1
599:SSL alert number 100
```

**Imagem 3.** Nessa imagem contem o erro obtido usando o código com a porta 465.

```
MAIL FROM:<bernardorizzone99@gmail.com>
250 2.1.0 OK d2e1a72fcca58-7a4140169b1sm124789b3a.5 - gsmt
RCPT TO:<bernardorizzone98@gmail.com>
RENEGOTIATING
40479E685F7B0000:error:0A00010A:SSL routines:can_renegotiate
:wrong ssl version:../ssl/ssl_lib.c:2323:
```

### Codigos e respostas completas (erros) :

```
rizzonly@DESKTOP-G55DHS5:~$ openssl s_client -connect smtp.gmail.com:465 -crlf
CONNECTED(00000003)
depth=2 C = US, O = Google Trust Services LLC, CN = GTS Root R1
verify return:1
depth=1 C = US, O = Google Trust Services, CN = WR2
verify return:1
depth=0 CN = smtp.gmail.com
verify return:1
---
Certificate chain
 0 s:CN = smtp.gmail.com
  i:C = US, O = Google Trust Services, CN = WR2
  a:PKEY: id-ecPublicKey, 256 (bit); sigalg: RSA-SHA256
  v:NotBefore: Oct  1 14:33:58 2025 GMT; NotAfter: Dec 24 14:33:57 2025 GMT
 1 s:C = US, O = Google Trust Services, CN = WR2
  i:C = US, O = Google Trust Services LLC, CN = GTS Root R1
  a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
  v:NotBefore: Dec 13 09:00:00 2023 GMT; NotAfter: Feb 20 14:00:00 2029 GMT
 2 s:C = US, O = Google Trust Services LLC, CN = GTS Root R1
  i:C = BE, O = GlobalSign nv-sa, OU = Root CA, CN = GlobalSign Root CA
  a:PKEY: rsaEncryption, 4096 (bit); sigalg: RSA-SHA256
  v:NotBefore: Jun 19 00:00:42 2020 GMT; NotAfter: Jan 28 00:00:42 2028 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIEVzCCAz+gAwIBAgIRAPpa4Pgwk+HEEMdZv1vQJkWDQYJKoZIhvcNAQELBQAw
OzELMAkGA1UEBhMCVVMxHjAcBgNVBAoTFUdvb2dsZSBUcnVzdCBTZXJ2aWNlc2EM
MAoGA1UEAxMDV1IyMB4XDTI1MTAwMTE0MzY1OTFoXDTI1MTIyNDU0MzY1OTFoGTEX
MBUGA1UEAxMOC210cC5nbWVpbC5jb20wWTATBgqhkiOQIBggqhkiOQMBBwNC
AATjCyGkbbLPGKc0dpXaGuQo0DjM4WhGtkeQjKInWkv0H7uU807Baci8uZknZg3
cZfVPvmnqSQJCj3NqkywmoVo4ICQTCCAaj0wDgYDVVR0PAQH/BAQDAgeAMBGA1Ud
JQQMMAoGCCsGAQUFBwMBMAwGA1UdEwEB/wQCMAAwHQYDVVR0BBYEFFq3gtMH7uCc
wJ9pgNxMTdc8HJjSMB8GA1UdIwQYMBaAFN4bHu15FdQ+NyTDIbvsNDltQrIwMFgG
CCsGAQUFBwEBBEwwSjAhBggrBgEFBQcwAYYVaHR0cDovL28ucGtpLmdvb2cvd3Iy
MCUGCCsGAQUFBzACHhloHRwOi8vaS5wa2kuZ29vZy93cjIuY3J0MBkGA1UdEQQS
MBCCDnNtdHAuZ21haWwUy29tMBMGA1UdIAQMMAowCAYGZ4EMAQIBMDYGA1UdHwQv
MC0wK6ApoCeGJWh0dHA6Ly9jLnBraS5nb29nL3dyMi9HU3LUMU40UEJyZy5jcmww
ggEEBgorBgEEAdZ5AgQCBIIH1BIHyAPAAdwAS8U40vVNyTIQGGcOPP3oT+0e1YoeI
nG0wBYTr5YYmOgAAAZmgaLXHAAAEAwBIMEYCIQC5wcqekBsMdTgSe09c7E9t2thS
i34PZEcsbBIdkbEuAIhAI2GcndL7feEdauGP2w5+nr/cj7VyICUoHs2f21NTDpN
AHUADeHymCvTDcFAYhIJ6LUu/Ed0fLHX6TDvDkIeth50qjQAAAGZoGi12AAABAMA
RjBEAiAIXObQYRsN78YlPQ683zsnhX/MH9CcGvF5waDJLGKCAAIgY6Nxdxyp7S3
WY93q+rmcj8YMo8J1t9y/wG1mGYt0BcwDQYJKoZIhvcNAQELBQADggEBAG6duMxQ
q/c/Ew7Ln5RQBFjiglr5erYV8FCvr+eNpnUtN0hmAyabM3lNlHL//y5Sde/eMAs1
bkuk7JBuRtG8Z6Y8lnp5kQt1hlnbFGIpqvcFluf+IsYlQN/4EKYy5CY70FqZDVwZ
1871ZV0eCEuEudj1+03CwKIFw0RupFMW6NLVVGWdQ62Hk/Uw8FScXBbMHlFRMCJT
Mr8qN2SlpUo4jhk4MxptmR3GF62d0CvSuc1HqyFdyumaa8qYe/OPf7CIeci5mXtA
OJ+poRPZppLD3n0jAcZ4Z+861mej6c/pkdyVF/cKoB1yFpNAT2FzHu8LXJRFU1k1
ZMoWloZRNns8UIY=
```

```
-----END CERTIFICATE-----
subject=CN = smtp.gmail.com
issuer=C = US, O = Google Trust Services, CN = WR2
---
No client certificate CA names sent
Peer signing digest: SHA256
Peer signature type: ECDSA
Server Temp Key: X25519, 253 bits
---
SSL handshake has read 4107 bytes and written 396 bytes
Verification: OK
---
New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384
Server public key is 256 bit
Secure Renegotiation IS NOT supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
Early data was not sent
Verify return code: 0 (ok)
---
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol : TLSv1.3
    Cipher : TLS_AES_256_GCM_SHA384
    Session-ID: 25EC794360CE9EA8E1575E672F017C9FCC8F70B0BE417FE5C1B1CCB08D1771D
    Session-ID-ctx:
    Resumption PSK: CBC4A9A08D910C12F14926C682667E1466299AA412CAB18365AEC38B5A4A675CCD45AF2CDAB
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 172800 (seconds)
    TLS session ticket:
0000 - 02 11 b1 6a cb 71 03 8e-26 a0 c6 0c 4f a6 58 a4 ...j.q...&...0.X.
0010 - fa 05 4d 6d 0d 07 c6 a6-c3 66 64 ce bc 28 ce 49 ..Mm.....fd..(.I
0020 - 02 9a 94 4a 5a 2e 68 6e-4e 25 aa 49 b9 1a f4 38 ...JZ.hnN%.I...8
0030 - 1a 11 89 95 9a d6 f7 3d-68 be 07 d4 8d 2c 2f f1 .....=h....,/.
0040 - f6 af 3d 97 cb 8d b4 83-b4 02 37 7f 14 3e db d8 ..=.....7...>...
0050 - 5c 25 93 4a 3b 92 0c 0f-e5 fe 46 5a cb e9 fa c2 \%.J;.....FZ....
0060 - 91 33 18 22 32 3a ee 7b-48 45 ed f6 ef 52 f9 31 .3."2:.{HE...R.1
0070 - 79 81 6b d8 be 9d f8 72-2b 7f 70 97 2c 9b 65 41 y.k....r+.p.,.eA
0080 - 32 cd 08 0b 72 31 ac e1-0a f7 d8 26 0f 9b 77 e6 2...r1.....&..w.
0090 - 09 51 41 3b 31 53 1a 8e-f5 80 93 cd 2b 27 ac 94 .QA;1S.....+'..
00a0 - 02 8e 3e 0b f4 4d 6d 59-05 83 88 30 e1 f2 cf 9c ..>..MmY...0....
00b0 - 15 8c 18 f0 e9 ca a7 34-53 c7 fc 99 a1 37 0a 69 .....4S.....7.i
00c0 - 8c 10 bf e2 ab 9d 2a 03-60 1e 9f 17 e1 72 6d 22 .....*.`.....rm"
00d0 - d4 b3 72 9e b0 d0 e2 0f-42 68 43 12 f0 78 7d 61 ..r.....BhC..x}a
00e0 - c8 73 15 0d 53 43 4c 38-f3 52 9d 54 c4 5f 19 ...s...SCL8.R.T._.

Start Time: 1761338673
```

```

Protocol : TLSv1.3
Cipher   : TLS_AES_256_GCM_SHA384
Session-ID: 4EEC9814C30FF8E7E76332E9BDB9348D7142CD38FBDDF1295E93EB857CFA1564
Session-ID-ctx:
Resumption PSK: D2B075F88EE5D68E4FDC0FC6712C43015D6BA8A6821CDCABB4AB0FD6A1C5D1FB38088D4CCD38F2A9F19A1
PSK identity: None
PSK identity hint: None
SRP username: None
TLS session ticket lifetime hint: 172800 (seconds)
TLS session ticket:
0000 - 02 11 b1 6a cb 71 03 8e-26 a0 c6 0c 4f a6 58 a4 ...j.q...&...O.X.
0010 - 58 78 46 3f 40 8d fd 91-4c 62 2c 4d 7a f3 05 fc XxF?@...Lb,Mz...
0020 - 03 20 66 bd 8d 54 f5 5d-a3 ac 86 f5 b9 92 19 0f . f..T.].....
0030 - d0 fd 1a 92 4f 2d fc cb-d5 98 f4 24 87 17 a2 b4 ....0-.....$.
0040 - fb 5c 86 f3 4c 9a ac 6c-47 d2 f4 ba 7e 7b ab 86 .\..L..lG...~{..
0050 - 08 dd 6e 6a 78 9c 16 77-a9 c1 db 63 06 05 d5 36 ..njx..w....c...6
0060 - fc b7 e9 f3 29 51 ee d3-88 34 9f a0 8e 7e 74 90 ....)Q...4...~t.
0070 - f4 0d 80 23 34 a3 9c 79-f5 9b f5 d1 3a fb 07 98 ...#4..y.....
0080 - 0d 32 65 a8 66 51 c7 63-50 3e 48 9f f9 7f 5a 42 .2e.fQ.cP>H...ZB
0090 - dd b7 91 e6 7a ff 66 d6-1e 7c a8 a7 5c 9a 91 db ....z.f..|..\...
00a0 - d7 bf a6 1b b6 57 cd 1f-69 31 72 c1 c3 7c 66 12 .....W..i1r..|f.
00b0 - fa 13 cf 16 94 b4 20 d5-b2 cc a8 c6 f3 ee 92 87 .....
00c0 - 85 27 57 46 05 9c f5 27-1e 0b b1 81 71 0f 47 ee . 'WF...'.....q.G.
00d0 - 4d ed 0d 14 6e 5b 7a b9-00 ac 0e 29 44 f8 73 51 M...n[z....)D.sQ
00e0 - 4d ce 18 9f 53 43 4c cd-c9 5e 29 31 e7 ad 0a M...SCL..^)1...

Start Time: 1761338673
Timeout : 7200 (sec)
Verify return code: 0 (ok)
Extended master secret: no
Max Early Data: 0
---
read R BLOCK
220 smtp.gmail.com ESMTP d2e1a72fcc58-7a4140169b1sm124789b3a.5 - smtp
EHLO google.com
250-smtp.gmail.com at your service, [189.6.37.55]
250-SIZE 35882577
250-8BITMIME
250-AUTH LOGIN PLAIN XOAUTH2 PLAIN-CLIENTTOKEN OAUTHBEARER XOAUTH
250-ENHANCEDSTATUSCODES
250-PIPELINING
250-CHUNKING
250-SMTPUTF8
AUTH LOGIN
334 VXNlcm5hbWU6
YmVybmFyZG9yaXp6b25lOTlAZ21haWwY29t
334 UGFzc3dvcmQ6
enBoZyBmc3J2IGFsYWogYW1tZg==
235 2.7.0 Accepted
MAIL FROM:<bernardorizzone99@gmail.com>
250 2.1.0 OK d2e1a72fcc58-7a4140169b1sm124789b3a.5 - smtp

```

```
flizzonly@DESKTOP-GSSDHSU: $ openssl s_client -starttls smtp -connect smtp.gmail.com:587 -crlf -tls1_2
CONNECTED(00000003)
depth=2 C = US, O = Google Trust Services LLC, CN = GTS Root R1
verify return:1
depth=1 C = US, O = Google Trust Services, CN = WR2
verify return:1
depth=0 CN = smtp.gmail.com
verify return:1
---
Certificate chain
 0 s:CN = smtp.gmail.com
  i:C = US, O = Google Trust Services, CN = WR2
  a:PKEY: id-ecPublicKey, 256 (bit); sigalg: RSA-SHA256
  v:NotBefore: Oct 1 14:33:58 2025 GMT; NotAfter: Dec 24 14:33:57 2025 GMT
 1 s:C = US, O = Google Trust Services, CN = WR2
  i:C = US, O = Google Trust Services LLC, CN = GTS Root R1
  a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
  v:NotBefore: Dec 13 09:00:00 2023 GMT; NotAfter: Feb 20 14:00:00 2029 GMT
 2 s:C = US, O = Google Trust Services LLC, CN = GTS Root R1
  i:C = BE, O = GlobalSign nv-sa, OU = Root CA, CN = GlobalSign Root CA
  a:PKEY: rsaEncryption, 4096 (bit); sigalg: RSA-SHA256
  v:NotBefore: Jun 19 00:00:42 2020 GMT; NotAfter: Jan 28 00:00:42 2028 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIEVzCCAz+gAwIBAgIRAPpa4Pgwk+HEEMdZv1vQJkwDQYJKoZIhvcNAQELBQAw
OzELMAkGA1UEBhMCVVMxHjAcBgNVBAoTFUdvb2dsZSBSUcnVzdCBTZXJ2aWNlc2EM
MAoGA1UEAxMDV11yMB4XDTI1MTAwMTE0MzY1MjE0MTEyNDE0MzY1MTEwGTEX
MBUGA1UEAxMOC210cC5nbWVpbC5jb20wWTATBgqhkhjOPQIBBggqhkhjOPQMBBwNC
AATjCygkbbLPgKc0dpXaGuQo0DJm4WhGtkeQjckInWkv0H7uU807Baci8uZknZg3
cZFVPvmnqSQCjJc3NQkywmoVo4ICQTCCAjdWdGyDVR0PAQH/BAQDAgeAMBMA1Ud
JQOMMAoGCCsGAQUFBwMBMAwGA1UdEwEB/wQCMAAwHQYDVR0OBBYEFFq3gtMH7uCc
wJ9pgNxmTdc8HJjSMB8GA1UdIwQYMBaAFN4bHu15FdQ+NyTDIbvsNDltQrIwMfG
CCsGAQUFBwEBBEwwSjAhBggrBgEFBQcwAYYVaHR0cDovL28ucGtpLmdvb2cvd3Iy
MCUGCCsGAQUFBzACHhloDHHRwOi8vaS5wa2kuZ29vZy93c3ciIuY3J0MBkGA1UdEQQ5
MBCCDnNtdHAuZ21haWwY29tMBMGA1UdIAQMMAAowCAYGZ4EMAQIBMDYGA1UdHwQv
MC0wK6ApoCeGJWh0dHA6Ly9jLnBraS5nb29nL3dyMl9HU3LUMU40UEJyZy5jcmww
ggEEBgorBgEEAdZ5AQCBIIH1BIHyAPAAAdAS8U40vVNYTIQGGcOPP3oT+0e1YoeI
nG0wBYTrSYymOgAAAZmgaLXHA4AAEAwBI MEYCIQC5wcqekBsMdTgSe09c7E9t2thS
i34PZEEsbBIDkbEuAIhAI2Gcndl7FeEduGP2w5+nr/cj7VyICu0Hs2f21NTDpN
AHUADeHymCvTdcFAYhIJ6LUu/Ed0fLHX6TDvDkIeth50qjQAAAGZoGi12AAABAMA
RjBEAIAIX0bQYRSn78YlpQ683zsnhX/MH9CcGvF5waDjLGKCAAIgY6Nxlxdxyp7S3
WY93q+rmcj8YMo8J1t9y/wG1mGYt0BcwDQYJKoZIhvcNAQELBQADggEBAG6duMxQ
q/c/Ew7Ln5RQBfjiglr5erYV8FCvr+eNpnUtN0hmAyabM3LnlHL//y5Sde/eMA51
bkuk7JBuRtG8Z6Y8lnp5kQt1hlnbFGIpqvcFlUf+IsYlQN/4EKYy5CY70FqZDVwZ
1871ZV0eCEuEudj1+03CwKIFw0RupFMW6NLVVGWdQ62Hk/Uw8FScXBbMHLfRMCJT
Mr8qN2SlpUo4jhk4MxptmR3GF62d0CvSuc1HqyFdyumaa8qYe/OPf7CIeci5mXtA
OJ+poRPZpplD3n0jAcZ4Z+861mej6c/pkdyVF/cKoB1yFpNAT2FzHu8LXJRFU1k1
ZMoWloZRNns8UIY=
-----END CERTIFICATE-----
```

```

subject=CN = smtp.gmail.com
issuer=C = US, O = Google Trust Services, CN = WR2
---
No client certificate CA names sent
Peer signing digest: SHA256
Peer signature type: ECDSA
Server Temp Key: X25519, 253 bits
---
SSL handshake has read 4566 bytes and written 329 bytes
Verification: OK
---
New, TLSv1.2, Cipher is ECDHE-ECDSA-CHACHA20-POLY1305
Server public key is 256 bit
Secure Renegotiation IS supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
SSL-Session:
    Protocol : TLSv1.2
    Cipher   : ECDHE-ECDSA-CHACHA20-POLY1305
    Session-ID: 8B781D07F15D40F5D3ADABCF86F2FFDD04D2EA4959F040D7968B37AC31E822A
    Session-ID-ctx:
    Master-Key: 797585A930EF41104B003C42C715DA2B732DFE89D31937806E50365082A588557A9404D6C731D78BCC07464D3F735A5
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 100800 (seconds)
    TLS session ticket:
0000 - 02 11 b1 6a cb 71 03 8e-26 a0 c6 0c 4f a6 58 a4    ...j.q..&...O.X.
0010 - cb 5f c1 c8 cb 95 72 b3-46 2e 5d fb 5d f6 28 c6    _.....r.F.].(.
0020 - 49 d3 e1 0e 93 b5 7d 55-7c b7 27 69 57 dd e7 65    I.....]U|. 'iW..e
0030 - 73 e5 38 72 dc 10 54 c9-29 f4 55 a9 ab 37 53 32    s.8r..T.).U..7S2
0040 - 79 f8 6f cb dc 86 94 23-9c d0 bb 1d 8e d3 46 30    y.o.....#.....F0
0050 - 1d f6 1c cd e8 fe ea a0-cc a5 1c 78 a1 28 a4 69    .....x.(.i
0060 - be 62 31 72 5d f1 5b 12-17 7f 9a 4d 7f 8f 51 f5    .b1r].[....M..Q.
0070 - 68 e1 74 03 28 50 7e 16-d4 ab 38 76 d2 af ba 1c    h.t.(P~...8v....
0080 - 9e 6f 92 13 94 08 b4 27-b4 55 43 6a 21 a3 d5 66    .o.....'UCj!..f
0090 - 4a 0c 41 77 f2 ff c6 c6-46 67 13 67 17 16 d7 fb    J.Aw....Fg.g....
00a0 - 01 91 70 95 8b cf e8 d7-53 4c 3f 7e a0 82 8f 8a    ..p.....SL?~....
00b0 - 17 a6 0f 7d aa ca b8 c4-93 e8 a7 bc 2b e3 8c 72    ...}......+...r
00c0 - bc b2 b5 50 b2 fc b4 6e-88 88 62 93 ac d7 28 ef    ...P...n..b...(.
00d0 - 7e 7f 34 77 9c 78 73 9a-ae 4e 53 43 4c c8 0c eb    ~.4w.xs..NSCL...
00e0 - 1d f1 4b a6 9a                                ..K..

    Start Time: 1761338576
    Timeout   : 7200 (sec)
    Verify return code: 0 (ok)
    Extended master secret: yes
---
250 SMTPUTF8
EHLO google.com

```

```

---
250 SMTPUTF8
EHLO google.com
250-smtp.gmail.com at your service, [189.6.37.55]
250-SIZE 35882577
250-8BITIME
250-AUTH LOGIN PLAIN XOAUTH2 PLAIN-CLIENTTOKEN OAUTHBEARER XOAUTH
250-ENHANCEDSTATUSCODES
250-PIPELINING
250-CHUNKING
250 SMTPUTF8
AUTH LOGIN
334 VXNlcm5hbWU6
YmVybmFyZG9yaXp6b25lOTLAZ21haWwY29t
334 UGFzc3dvcmQ6
enBoZyBnc3J2IGFsYWogYWltZg==
235 2.7.0 Accepted
MAIL FROM:<bernardorizzone99@gmail.com>
250 2.1.0 OK 98e67ed59e1d1-33fed73a7b7sm42733a91.5 - gsmt
RCPT TO:<bernardorizzone99@gmail.com>
RENEGOTIATING
4007DD2988790000:error:0A00044C:SSL routines:ssl3_read_bytes:tlsv1 alert no renegotiation:../ssl/record/rec_layer_s3.c:1599:SSL alert number 100

```

**Exito obtido pelo uso da ferramenta swaks:**



